

DRUMPANEL · REQUIREMENTS

DrumPanel requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/drumpanel		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-03	Amish Chadha	Round-2 decision 5A (DMP-DDR-004): R7 kept at 2.5 drums an hour; the TRL 4 timed trial decides; no design change
0.4	2026-10-03	Amish Chadha	R7 restated to 2.5 drums an hour by Amish on 2026-10-03 (decision 5A, DMP-DDR-003); status of R5, R6, R7, R9 and R11 from DMP-CAL-001 v0.2 after the notching punch, ring head and dial adjuster
0.3	2026-10-03	Amish Chadha	TRL 3; status of every requirement from DMP-CAL-001 on the constructable design (DMP-DDR-002); R9 worded against the value-engineering target; R12 changed to three panels per drum
0.2	2026-10-03	Amish Chadha	TRL 2; targets firmed up; R10 guarding, R11 handling and R12 yield added; concept status for each requirement
0.1	2026-09-30	Amish Chadha	Initial scaffold

DrumPanel requirements

On paper, the constructable design meets ten of the twelve requirements, misses one, and is over the value-engineering target of R9. On 2026-10-03 Amish chose option A on every requirement decision put to him ("1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"); for DrumPanel that added a lever notching punch and a ring head and restated R7 from 4 to 2.5 drums an hour (5A), and added a fine-pitch dial adjuster to the bending roll (8A) (DMP-DDR-003). R7 is still not met: the estimate is about 2.26 drums an hour, up from 2.0. R5 is now met on paper with a trial pass for each batch of drum steel: the dial sets the bending roll to 0.025 mm against a 0.29 mm allowance. Every figure is from DMP-CAL-001 v0.2 (docs/04-calcs/sizing.py); the tag in brackets is the line of its output.

Table 1. Requirements and their status at TRL 3.

ID	Requirement	Target	Verification (TRL 4)	Status at TRL 3
R1	No open flame in the process	Zero heat or flame steps from drum to flat sheet	Procedure review and observed trial	Met by design: purge, check, cut and roll are all cold; no grinder or torch touches a drum
R2	Vapour check before cutting	Check reliably flags a drum with flammable vapour at or above 10 % of the lower explosive limit (LEL)	Bump test of the bought detector with 50 % LEL gas; bench test with a calibrated gas	Met by design: a bought pumped LEL detector with its alarm set at 5 % LEL, half the target (DMP-DDR-001, D2)
R3	Remove ends cold	Both heads off a 208 L drum in under 10 minutes by one person	Timed trials on a batch of drums	Met on paper: about 3.9 min [H7]; crank force 12 to 27 N [C10 to C15]
R4	Open the side	Straight slit along the full body length, edge deviation at or below 5 mm	Measure the cut line on trial drums	Met by design: the shear head runs on a rail straight within 1 mm
R5	Flat sheet	Panels flat within 10 mm over 1 m after rolling	Straightedge and feeler check	Met on paper, with a trial pass for each batch: needs the bending roll within 0.29 mm of the right height [E13], which is 11.8 divisions of the new 0.025 mm dial [E14d], held by a lock nut; a fixed setting can leave 188 mm of sag on other steel [E12], so the reading is found by a trial pass on the first drum of each batch; the last 90 mm of each panel end is set by hand [E15, E16]
R6	Manageable effort	Crank force at or below 150 N at the handle	Spring scale on the handle during trials	Met on paper: end cutter 27 N, shear head 68 N for the slit and 136 N for two ring cuts at 1.5 mm wall, notching punch lever 131 N, slip roll 20 N [C15, D15, D15r, K3, E23]
R7	Throughput	At least 2.5 drums to flat sheet per hour for a two-person team (restated 2026-10-03, was 4)	Timed half-day workshop trial	Not met on paper: about 2.26 drums an hour [H5], 53.1 hands-on minutes a drum against 48.0 [H4, H9]. Kept at 2.5 by Amish (round 2); the TRL 4 timed trial decides
R8	Local build	All parts made or bought in a town with a welder and a lathe	Build by a partner fabricator from the drawings	Met by design: welding, drilling and turning only; the rolls need a lathe with 1.1 m between centres; discs, cutter wheel, bearings, gears and chain are bought
R9	Value engineering	Bench set parts cost at or below the USD 3,000 value-engineering target	Costed bill of materials	Over the target: estimated USD 3,489, USD 489 over the value-engineering target [J1 to J3]
R10	Guarding	No finger reaches a roll nip, disc nip, gear or chain in normal use; guards need a tool to remove	ISO 13857 check of the guards as built	Met on paper: every slot 8 mm or less and 45 mm or more from its nip [G1 to G3]; the shear head's chains run inside its closed drive case and the ring head coupling turns inside a guard sleeve

ID	Requirement	Target	Verification (TRL 4)	Status at TRL 3
R11	Handling	No lift over 25 kg by one person; a drum full of water is never lifted	Weigh parts; procedure review	Met with a two-person rule: heaviest piece the 34 kg cradle frame [I4]; shear head 23.7 kg, ring head 20.4 kg, notching punch 18.7 kg [I5, I7, I9]; a full drum (about 240 kg) is filled and drained in place [B2]
R12	Yield	Per drum: three flat panels about 233 x 1,800 mm and two head discs about 550 mm across	Measure the output of trial drums	Met by design: panels 233, 234 and 233 x 1,800 mm; heads 552 mm [A1 to A3]

■ Requirement restated on 2026-10-03

R7 was "at least 4 drums to flat sheet per hour for a two-person team". Amish restated it to 2.5 drums an hour when he chose option 5A on 2026-10-03 ("1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"): add a lever notching punch and a second shear head so two ring cuts are made per pass, and restate R7 to 2.5 drums an hour. The record is DMP-DDR-003.

In round 2 Amish kept R7 at 2.5 drums an hour (5A of DMP-DDR-004): "i agree with all the 46 recommendations you provided. please proceed." The paper estimate of 2.26 drums an hour is not changed, and R7 is not claimed as met. The gap is about 10 % of the hands-on time, inside the uncertainty of the task-time estimates, so the TRL 4 timed half-day trial decides, and no design change is made at TRL 3.

■ Assumptions

- Most drums reaching artisans held oils or similar contents that a water purge can clear enough to cut cold. Drums that did not are rejected at the door (DMP-PRB-001, Constraints).
- Artists accept painted panels, with paint removal as a separate cold step (DMP-DDR-001, D4).
- Drum body steel has a yield strength of 200 to 300 MPa and a wall of 0.8 to 1.2 mm; heads are 1.0 to 1.5 mm.
- The rolling hoops are cut out as two strips, so the slip roll never has to straighten them (DMP-DDR-002, P7).
- A pumped LEL detector, bump-tested each day, is good enough for a pass or fail check before cold cutting.

■ Safety

Safety: R1, R2, R10 and R11 are safety requirements. They are not traded against cost or throughput: where a cheaper or faster option would weaken them, the conservative option was taken (DMP-DEC-001). The bench set is an open engineering reference, never certified equipment.